

Original article

Genomic features and pathophysiological impact of a multidrug-resistant *Staphylococcus warneri* variant in murine mastitisM. Nazmul Hoque^{a,*}, Golam Mahbub Faisal^a, Ziban Chandra Das^a, Tahsin Islam Sakif^b, Mamun Al Mahtab^c, M. Anwar Hossain^d, Tofazzal Islam^e^a Molecular Biology and Bioinformatics Laboratory, Department of Gynecology, Obstetrics and Reproductive Health, Bangabandhu Sheikh Mujibur Rahman Agricultural University (BSMRAU), Gazipur 1706, Bangladesh^b Keck Graduate Institute, 535 Watson Drive, Claremont, CA 91711, USA^c Interventional Hepatology Division, Bangabandhu Sheikh Mujib Medical University, Dhaka 1000, Bangladesh^d Jashore University of Science and Technology, Jashore 7408, Bangladesh^e Institute of Biotechnology and Genetic Engineering (IBGE), BSMRAU, Gazipur 1706, Bangladesh

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ABSTRACT

Non-aureus staphylococci (NAS) represent a major etiological agent in dairy animal mastitis, yet their role and impact remain insufficiently studied. This study aimed to elucidate the genomic characteristics of a newly identified multidrug-resistant NAS strain, specifically *Staphylococcus warneri* G1M1F, isolated from murine feces in an experimental mastitis model. Surprisingly, NAS species accounted for 54.35 % of murine mastitis cases, with *S. warneri* being the most prevalent at 40.0 %. *S. warneri* G1M1F exhibited resistance to 10 major antibiotics. Whole-genome sequencing established a genetic connection between G1M1F and *S. warneri* strains isolated previously from various sources including mastitis milk in dairy animals, human feces and blood across diverse geographical regions. Genomic analysis of *S. warneri* G1M1F unveiled 34 antimicrobial resistance genes (ARGs), 30 virulence factor genes (VFGs), and 278 metabolic features. A significant portion of identified ARGs (64 %) conferred resistance through antibiotic efflux pumps, while VFGs primarily related to bacterial adherence and biofilm formation. Inoculation with G1M1F in mice resulted in pronounced inflammatory lesions in mammary and colon tissues, indicating pathogenic potential. Our findings highlight distinctive genomic traits in *S. warneri* G1M1F, signifying the emergence of a novel multidrug-resistant NAS variant. These insights contribute to understanding NAS-related mastitis pathophysiology and inform strategies for effective treatment in dairy animals.

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1. Background

Staphylococci are the primary etiological agents of mastitis in dairy animals [1,2], representing a major global threat and economic burden for dairy farmers [3]. The genus *Staphylococcus* comprises around 85 species and 30 subspecies [4]. In bovine mastitis, staphylococci are classified into two groups: one that

includes *Staphylococcus aureus*, one of the major pathogens, and a second group comprises other staphylococci, mostly the minor pathogens and popularly termed the coagulase-negative staphylococci (CoNS) [5]. Recently, a new classification scheme has been introduced in mastitis literature that involves grouping all staphylococci other than *S. aureus* into a single category, non-aureus staphylococci (NAS) or non-aureus staphylococci and mammaliococci (NASM) [6]. These organisms, which include various species within the *Staphylococcus/Mammaliococcus* genus apart from *S. aureus*, can lead to inflammation of the udder in cows, goats, and other domestic ruminants, impacting milk production and animal health [7]. However, the ecology, epidemiology, effect on udder health, and virulence or protective traits of the mastitis associated NAS or NASM species vary in different host species [6–8]. Several studies have reported that *Staphylococcus chromogenes*,

Abbreviations: NAS, non-aureus staphylococci; NASM, non-aureus staphylococci and mammaliococci; *S. warneri*, *Staphylococcus warneri*; MDR, multidrug-Resistant; MIC, minimum inhibitory concentration; PBS, phosphate buffered saline; WGS, whole-genome sequencing; ML, maximum likelihood; CDS, coding sequences; ANI, average nucleotide identity; ARGs, antimicrobial resistance genes; VFGs, virulence factor genes.

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Staphylococcus epidermidis, *Staphylococcus xylosus*, *Staphylococcus vitulinus*, *Staphylococcus simulans*, *Staphylococcus warneri*, and *Mammaliococcus sciuri* are the main NAS species causing mastitis in dairy animals [6–9]. *M. sciuri*, previously known as *Staphylococcus sciuri*, is considered one of the oldest species within the evolutionary history of the *Staphylococcus/Mammaliococcus* genus [10]. Some of the NAS species are considered true mastitis pathogens with potential virulence factors, a high level of resistance to multiple antimicrobials, and the ability to cause chronic infections [11,12]. Moreover, the emergence of multidrug-Resistant (MDR) strains among NAS species has become a global public health concern [4,9,13]. The *S. warneri* is an opportunistic NAS pathogen that is capable of causing several infections, including mastitis [4,9,13]. Although the role of *S. aureus* as an etiological agent of bovine mastitis has been elucidated previously [1,2,14], studies on the relative importance of specific NAS species, for example, *S. warneri*, in mastitis have not been undertaken in Bangladesh.

The emergence of MDR NAS species, such as *S. warneri*, presents a concerning trend due to their potential impact on dairy industries and public health. These bacteria can exhibit resistance to multiple antibiotics, making treatment more challenging and raising concerns about the spread of antibiotic resistance. Therefore, vigilant surveillance, prudent use of antibiotics, and robust infection control measures are essential to address the threat posed by MDR bacterial species. Indeed, the clinical significance and pathogenic potential of *S. warneri*, are still subjects of ongoing research and debate. While some NAS species have been identified as pathogens associated with mastitis in domestic ruminants and have shown certain virulence factors [15,16], their role in infections and their overall pathogenicity compared to more well-known pathogens like *S. aureus* are areas of active investigation. Continued research is necessary to better understand the pathogenic potential and clinical relevance of these emerging MDR NAS species. Traditional phenotypic methods have been widely utilized for monitoring and managing the spread of antibiotic-resistant bacterial pathogens in mastitis across the globe [1,2]. Phenotypic methods involve culturing the bacteria and then subjecting them to various tests to determine their susceptibility to antibiotics. These tests help to identify the antibiotics to which the bacteria are resistant or sensitive [2]. However, these conventional approaches often fall short in distinguishing closely related strains and detecting virulence or resistance traits. Molecular techniques such as polymerase chain reaction (PCR), whole-genome sequencing (WGS), and other molecular assays are used to detect specific ARGs, mutations, or genetic markers associated with antibiotic resistance in bacterial strains. Molecular methods offer high specificity and can detect resistance mechanisms more rapidly than traditional culture-based methods [17]. Moreover, DNA sequence-based species identification is currently considered the gold standard for identifying NAS species with precision [4,6]. Recent advancements in next-generation sequencing technologies, such as whole-genome sequencing (WGS), have effectively addressed the limitations of the phenotypic methods and can encompass all pertinent genomic attributes. The cutting edge WGS methods not only assist in species identification but also aid in understanding the genetic diversity, virulence factors, and antibiotic resistance profiles of different NAS species, providing crucial information for managing mastitis and tracking their spread and impact in various settings [15,16]. Due to increasing concerns about emerging and opportunistic pathogens in dairy animal mastitis [6–9], our goal was to investigate the genomic capabilities of *S. warneri* strain G1M1F. This involved a detailed examination of antimicrobial resistance, virulence factors, and metabolic functions through comprehensive whole-genome sequencing. The *S. warneri* G1M1F was isolated from the gut (feces) of a mouse with experimentally induced mastitis.

2. Materials and methods

2.1. Ethical statement

The Animal Research Ethics Committee (AREC) of the Bangabandhu Sheikh Mujibur Rahman Agricultural University, Bangladesh, reviewed and approved the experimental procedures of this study (Reference number: FVMAS/AREC/2023/6679, Date: 16/01/2023).

2.2. Non-aureus staphylococcal isolates and their antimicrobial susceptibility

A cow-to-mouse mastitis model was established (at our laboratory animal research facility, 24.09°N, 90.41°E) through transplantation of fecal or milk microbiota of cow (clinical mastitis cow) to germ-free (GF) timed-pregnant Swiss albino mice (N = 52) following our previously published protocol [18]. In brief, the challenged mice received 1 g/kg fecal suspension (Group – I; n = 21) or 0.5 mL of milk suspension (Group – II; n = 21) orally at Day 17 of their gestation while control group mice (Group – III; n = 10) received a placebo with distilled water (0.5 mL/mice, orally). The mice were observed for mastitis syndrome (e.g., swollen, red and inflamed mammary glands) development, and sacrificed 10 days after microbiota transplantation (at Day 27 of their gestation) for sample collection [18]. Fecal samples (N = 42) from these experimentally induced mastitis mice were collected, homogenized (1:1) in PBS (Phosphate-buffered saline) and serially diluted (1:10). Dilutions were plated onto nutrient agar plates and incubated at 37 °C for 24 h. Pure colonies were isolated and subsequently streaked on mannitol salt agar (Oxoid, Thermo Scientific, UK) plates for incubation at 37 °C for 24 h [14]. Colonies suspected of being staphylococci (46 isolates) were initially identified by their colony morphology, Gram staining (Gram-positive cocci), and biochemical tests such as catalase, indole, methyl red, Voges-Proskauer (VP), oxidase, urease, triple sugar iron, and coagulase tests [17]. The species-level identification of NAS (25 isolates) was performed through a VITEK-2 system (version 9.01). From these NAS species, 10 isolates were detected as *S. warneri* [19]. Antimicrobial susceptibility patterns of the confirmed NAS isolates (n = 25) were examined using the disk diffusion method following the guidelines of the Clinical Laboratory Standards Institute (CLSI) M100 33rd Edition [20]. Antibiotics were selected for susceptibility testing corresponding to a panel of antimicrobial agents (CM0337, Oxoid™, Thermo Scientific, UK) of interest to the dairy industry and public health, and commonly used by veterinary practitioners in Bangladesh. The groups of antimicrobials used were - Beta-lactams (ampicillin, 10 µg/mL; oxacillin, 1 µg/mL), Tetracyclines (doxycycline, 30 µg/mL; tetracycline, 30 µg/mL), Nitrofurans (nitrofurantoin, 300 µg/mL), Fluoroquinolones (ciprofloxacin, 10 µg/mL; nalidixic acid, 30 µg/mL), Cephalosporins (cefoxitin, 30 µg/mL), Carbapenems (imipenem, 10 µg/mL), Aminoglycosides (gentamycin, 10 µg/mL; streptomycin, 10 µg/mL), Chloramphenicol (chloramphenicol, 30 µg/mL), Macrolides (azithromycin, 15 µg/mL), and Sulphonamides (sulphonamide, 300 µg/mL). Resistance was defined according to CLSI guidelines [20]. We further determined the minimum inhibitory concentration (MIC) values of MDR (resistant to ≥ 3 antibiotics) isolates (n = 17) against commonly used antibiotics according to the broth dilution method recommended by CLSI [20]. Based on antimicrobial susceptibility results, we selected one multidrug resistant isolate (*S. warneri* G1M1F; one of the isolates from Group – I mice) that showed resistance against the highest number (n = 10) of antibiotics. The *S. warneri* G1M1F was further confirmed by 16S rRNA gene sequencing.

2.3. Complete genome sequencing, assembly and annotation of *S. warneri* G1M1F

The genomic DNA from one of the NAS isolates (e.g., *S. warneri* G1M1F), with the highest MDR pattern (resistance to 10 antibiotics), was extracted using the QIAamp DNA minikit (Qiagen, Hilden, Germany), following the manufacturer's instructions. The extracted DNA was checked for quality using agarose gel electrophoresis and quantified using the NanoDrop 2000 UV-visible (UV-Vis) spectrophotometer (Thermo Fisher, Waltham, MA, USA). The Nextera DNA Flex library prep kit (Illumina, San Diego, USA) was used to generate libraries from 1 ng DNA, and whole-genome sequencing of the prepared libraries was performed using the Illumina MiSeq sequencer with a 2 × 250-bp protocol. The paired-end raw reads ($n = 38,002,288$) were trimmed using Trimmomatic v0.39 [21] to remove Illumina adapters, known Illumina artifacts, and phiX reads and quality checked using FastQC v0.11.7 (Babraham Bioinformatics). Reads with Phred scores of >20 were *de novo* assembled using SPAdes v3.15.5 [22], and the quality of the assembled genome was checked using QUAST v5.0.2 [23]. Genome completeness was assessed through CheckM v1.2.2 using the *Staphylococcus* CheckM marker set [24]. The NCBI Prokaryotic Genome Annotation Pipeline (PGAP) v6.4 (https://www.ncbi.nlm.nih.gov/genome/annotation_prok/) was used for genome annotation. PlasmidFinder [25], CRISPRimmunity (<http://www.microbiome-bigdata.com/CRISPRimmunity/index/home>) and PHAge Search Tool Enhanced Release; PHASTER server (<http://phaster.ca/>) were used to predict plasmid replicons, CRISPR (clustered regularly interspaced short palindromic repeats) arrays and phage-associated genes and genomic regions, respectively.

2.4. Phylogenetic analysis and genomic comparison of *S. warneri* G1M1F

To elucidate the genetic relatedness between G1M1F and 27 reference *S. warneri* genomes, retrieved from the National Center for Biotechnology Information (NCBI) GenBank database (Table S1), a maximum-likelihood (ML) tree was constructed. Genomes were aligned with MUSCLE v5.0 (<https://github.com/rcedgar/muscle>), and the ML phylogenetic tree was created using PhyML v3.0 [26], and finally visualized through iTOL (v3.5.4) [27]. Subsequently, the pairwise average nucleotide identity (ANI) of the *S. warneri* G1M1F and nine closely related *S. warneri* genomes (based on phylogenetic results) was computed using the FastANI calculator available in the GTDB-tk v2 database [28]. In addition, multiple genomes ($n = 7$, based on ANI results) alignment and visualization were performed using BLAST Ring Image Generator (BRIG) v0.95 [29], with *S. warneri* strain AM100_PB1_HT_1D_2A (GenBank Accession No. NZ_JAQLDX000000000) as reference (Table S1). Based on the results of the phylogenetic, ANI and BRIG analyses, Roary v3.11.2 [30] was used to perform pan-genome analysis. Through pan-genome analysis four different classes of genes such as 'core' (99 % ≤ strains ≤ 100 %), 'soft core' (95 % ≤ strains < 99 %), 'shell' (15 % ≤ strains < 95 %) and 'cloud' (0 % ≤ strains < 15 %) were predicted.

2.5. Genomic functional potentials analysis of *S. warneri* G1M1F

The assembled draft genome of G1M1F was further annotated to elucidate the genomic functional potentials such as antimicrobial resistance genes (ARGs), virulence factor genes (VFGs), and metabolic features. The ABRicate v1.0.1 (<https://github.com/tseemann/abricate>) bundled with multiple databases: NCBI AMRFinderPlus [31], CARD 2023 [32], ARG-ANNOT [33], and MEGARes 2.0 [34] was used to predict ARGs in G1M1F genome. The ARGs selection criteria were set to perfect (100 % identity) and strict (>95 % identity) hits

only to the curated reference sequences in the databases. The VFGs in the G1M1F genome (with 90 % nucleotide identity and query coverage) were identified using the VFDB v6.0 database [35] bundled in the ABRicate v1.0.1. The draft genome was also annotated using the RAST (Rapid Annotation using Subsystem Technology) server, v2.0 [36], to identify metabolic function related genes/pathways under different subsystem categories. Secondary metabolites in the study genomes were predicted using the antiSMASH v3.0 database [37].

2.6. Mouse mastitis model of infection with *S. warneri* G1M1F

To elucidate the virulence potentials of the *S. warneri* G1M1F, we further conducted a well-characterized and validated mouse mastitis model experiment [38]. In this model experimental challenge, twenty ($n = 20$) timed-pregnant (at Day 18 of breeding) GF Swiss albino mice were purchased from the International Centre for Diarrhoeal Disease Research, Bangladesh (ICDDR'B), Dhaka, Bangladesh. The mice were randomly divided into three groups: Group-I ($n = 7$), Group-II ($n = 7$), and Group-III ($n = 6$). Group-I mice were challenged with 100 µL of an overnight culture of *S. warneri* G1M1F (approximately 1×10^6 CFU/mL) by subcutaneous (s/c) injection in the inguinal regions [38], while Group-II mice inoculated with 100 µL overnight culture of *S. warneri* G1M1F (approximately 1×10^6 CFU/mL) orally, at Day 18 of their gestation. The control group mice (Group-III) received only a placebo with 100 µL distilled water (DW) orally. Syringes with 33-gauge blunt needles were used for administering the inocula. The mice were housed in a GF environment on a 12 h light/dark cycle with unlimited access to food and water. The mammary glands of mice were examined daily for the development of mastitis. The gross clinical signs of mastitis observed were redness, swelling and discoloration of mammary gland, and extrusion of exudate from the mammary glands (inguinal glands). At the end of the experiment (i.e., 10 days after challenge), mice were humanely sacrificed, mammary tissues and gut (colon) tissue were collected for estimating the level of inflammation and histological changes [38]. Histopathological procedures for preparing sections (hematoxylin and eosin stained) from mammary and colon tissue followed our previously published protocols [18]. The histopathological changes observed in mammary glands and colon of mice infected with *S. warneri* G1M1F, were graded on a scale of 0–3 (normal 0, mild 1, moderate 2, severe 3) [18,39]. The slides were assessed under a Primo Star (Carl Zeiss, Germany) microscope (40 × objective), and finally images were collected using an AxioCamERc5s camera through ZEN 2012 INK software (Carl Zeiss, Germany) and visualized using Image J software (<https://imagej.nih.gov/ij/>).

3. Results

3.1. Non-aureus staphylococci (NAS) isolates from the gut of mastitis induced mice show antibiotic resistance

Forty-six ($n = 4$) staphylococcal isolates were screened from 42 fecal samples collected from experimentally induced mastitis mice, of which 25 (54.35 %) isolates were identified as NAS. Among these NAS isolates, 10 (40.0 %) were identified as *S. warneri*, and other NAS species were *Staphylococcus chromogens* (28.0 %), *S. epidermidis* (24.0 %) and *S. simulans* (8.0 %). Antimicrobial susceptibility testing for the 25 NAS isolates showed that 68.0 % were MDR isolates, with the highest resistance rate against oxacillin (92.0 %), followed by sulphonamide (76.0 %), ampicillin (36.0 %), nalidixic acid (32.0 %) and gentamicin (28.0 %) (Fig. 1). The G1M1F strain exhibited resistance to 10 out of the 14 tested antibiotics, including nalidixic acid, nitrofurantoin, gentamicin, oxacillin, azithromycin,

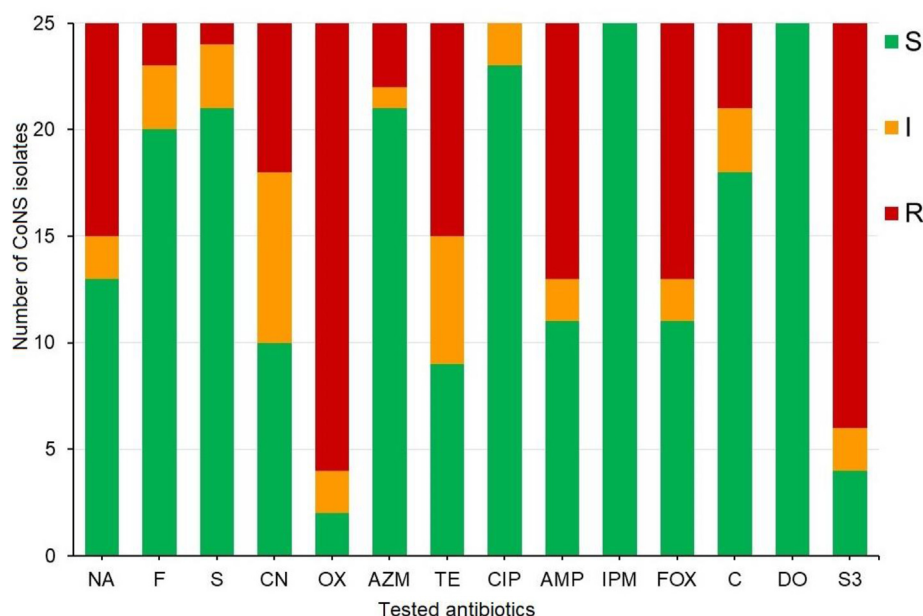


Fig. 1. Antimicrobial resistance profile of *S. warneri* isolated from mice with induced mastitis. NA, Nalidixic acid; F, Nitrofurantoin; S, Streptomycin; CN, Gentamicin; OX, Oxacillin; AZM, Azithromycin; TE, Tetracycline; CIP, Ciprofloxacin; AMP, Ampicillin; IPM, Imipenem; FOX, Cefoxitin; C, Chloramphenicol; DO, Doxycycline; S, Sulphonamide. R: resistance, I: intermediate resistance, and S: susceptible.

tetracycline, ampicillin, cefoxitin, chloramphenicol, and sulphonamide. However, this NAS isolate showed susceptibility to streptomycin, ciprofloxacin, imipenem, and doxycycline (Fig. 1, Table 1).

3.2. Genomic features and phylogenetic inference of *S. warneri* G1M1F

We sequenced the genome of *S. warneri* G1M1F, which revealed the highest MDR pattern (resistance to 10 antibiotics) among all of the NAS isolates. The draft genome of *S. warneri* G1M1F is composed of a circular chromosome spanning 2,504,515 base pairs (bp) with 32 contigs, 32.5 % GC content, and 99.64 % genome completeness (Fig. S1). It contains 2502 genes, with 2426 protein-coding genes and 38 RNA genes, including 31 tRNA, three rRNA and four ncRNA genes (Table S2). The SEED subsystems analysis performed using RAST, 1165 genes were assigned to 24 different functional categories and 278 subsystems. Two plasmid replicons such as rep5b and repUS35 of 28,679 and 8615 bp, respectively, were identified in the G1M1F genome (with 95 %

identity and 60 % coverage). In addition, two CRISPR arrays (e.g., Cas-TypeIIIA cluster and Cas-TypeIIU cluster) and two prophage regions (e.g., PHAGE_Staphy_vB_SauS_phi2_NC_028862-13 and PHAGE_Staphy_PT1028_NC_007045-13) were also predicted in this genome of *S. warneri* (Table S2). The maximum-likelihood tree showed seven major clades of 28 isolates (Table S1) from five continents (Fig. 2). Phylogenetic inference revealed the closest evolutionary relationship of *S. warneri* G1M1F (JAQPCE000000000_MM_BD) with another *S. warneri* strain isolated from human feces in China (JAQDLX010000000_HF_CHN). The G1M1F strain also showed a close evolutionary relationship with two *S. warneri* strains isolated from human blood (CP061041_Hum_CHN) and mouse feces (QXWP010000000_MC_CHN) in China (Fig. 2). Besides these genomes, the G1M1F strain had an evolutionary relationship with several *S. warneri* strains causing mastitis in cows (QXUP000000000_BM_CAN, QXUO000000000_BM_CAN, and JAHCPD000000000_BM_BEL), goat (CP049802_CM_BRA) and buffalo cow (CP031269_BuM_BRA) throughout the world (Fig. 2). The evolutionary phylogenetic inference of the *S. warneri* G1M1F strain was further validated by genomic similarity (ANI) analysis between closely related genomes. The ANI analysis results demonstrated that the *S. warneri* G1M1F exhibited an ANI value close to 100 % (Fig. S2) when compared to the reference genomes. This finding suggests that G1M1F is classified within the same species according to microbial taxonomy standards, as the ANI value meets the threshold (>95 % cut-off for ANI) for species delineation [28]. To our knowledge, this is the first WGS report of a NAS strain (e.g., *S. warneri* strain G1M1F) isolated from feces of a mouse with experimentally induced mastitis in Bangladesh.

In addition, a comparative genomic analysis was performed among closely related *S. warneri* strains (based on both phylogenetic and ANI results) through BRIG v0.95, where *S. warneri* SNUC 4113 (GenBank accession: QXUP000000000) was used as a reference. The circular genomic map from the BRIG illustration revealed a high level of sequence conservation among members of the *S. warneri* lineages. However, a significant number of non-homologous regions were found around the reference genome with over 95 % identity (Fig. 3). The G1M1F strain was found to

Table 1
Antibiotic susceptibility of non-aureus staphylococci (NAS) isolates screened from feces of mice with experimentally induced mastitis.

Antibiotic class	Antimicrobial	MIC ₉₀ (μg/ml)	SIR
Aminoglycosides	Gentamicin	<12	R
	Streptomycin	≥15	S
Carbapenems	Imipenem	>23	S
Beta-lactams	Ampicillin	<13	R
	Oxacillin	<10	R
Chloramphenicols	Chloramphenicol	<12	S
Cephalosporins	Cefoxitin	>14	R
Tetracyclines	Tetracycline	<14	R
	Doxycycline	<10	S
Fluoroquinolones	Ciprofloxacin	>21	S
	Nalidixic acid	13	R
Macrolides	Azithromycin	<13	R
Nitrofurans	Nitrofurantoin	<14	R
Sulphonamides	Sulphonamide	≤12	R

MIC: Minimum Inhibitory Concentration; S: Sensitive; I: Intermediate; R: Resistant.

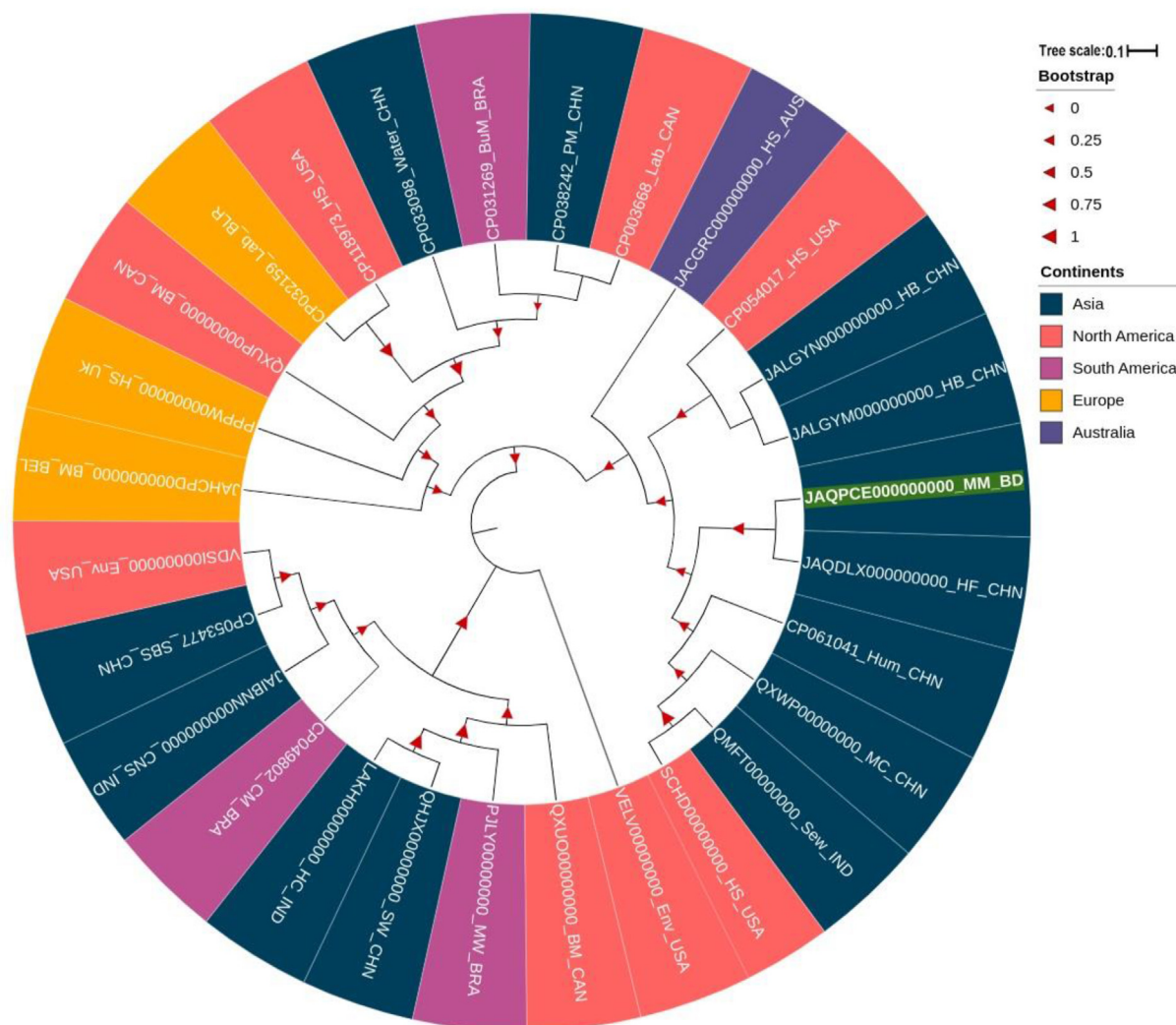


Fig. 2. The evolutionary phylogenetic relationships between *S. warneri* G1M1F and other *S. warneri* strains obtained from five continents of the world. Whole genome sequences of twenty-seven *S. warneri* strains retrieved from NCBI were used for phylogenetic analysis. The mid-point rooted tree was constructed using the NCBI Tree Viewer (<https://www.ncbi.nlm.nih.gov/tools/treeweb/>) and visualized with iTOL (Interactive Tree Of Life). The evolutionary relationship was inferred using the maximum-likelihood method. Different colors (e.g., dark pastel blue for Asia, pale red for North America, berry boost for South America, chrome yellow for Europe, and soft navy blue for Australia) are assigned according to the close evolutionary relatedness (clade) of the genomes. The scale bar is in the unit of the number of substitutions per site. The values on the branches are bootstrap support values based on 1000 replications. All the sequences were indicated by their accession numbers, followed by the host and country code. The country codes according to the standard abbreviation are: China (CHN), United States of America (USA), Australia (AUS), Canada (CAN), Brazil (BRA), Belarus (BLR), United Kingdom (UK), Belgium (BEL), India (IND) and Bangladesh (BD). The genome of the *S. warneri* strain G1M1F (JAQPCE000000000_MM_BD) is highlighted on a green background and white color fonts.

share common regions of genetic variations with the reference strain (SNUC 4113), mastitis causing *S. warneri* strains of goat (Cap_100.1), buffalo cow (16A), and dairy cow (Ani-LG-025), and two other *S. warneri* strains isolated from human feces (WS479, AM100_PB1_HT_1D_2A), at different sites on the genomes (Fig. 3). We further performed pan-genome analysis to better elucidate the differences in virulence genes of study genome and seven phylogenetically close reference strains of *S. warneri* (including G1M1F). The pan-genome matrix based on the presence and absence of genes in eight valid strains showed clusters of genes and dendrogram of the closely related *S. warneri* strains (Fig. 4a). In the pan-genome dendrogram, *S. warneri* strain Ani-LG-025 isolated from bovine mastitis milk was found as the closest relative of G1M1F strain (Fig. 4a). A total of 3420 genes were estimated in the pan-genomes (Fig. 4b). Among these, 1999 core genes were identified which were present in >99 % sequences of the valid strains. We also

detected 713 shell genes and 708 cloud genes, which were present in 15 %–95 % and <15 % of genomes, respectively. However, no soft-core gene was found in the pan-genome analysis (Fig. 4b).

3.3. Antimicrobial resistance, virulence, and metabolic potentials of *S. warneri* G1M1F

To elucidate the genomic potentials of *S. warneri* G1M1F, we comprehensively analysed the ARGs, VFGs, metabolic features, and secondary metabolites. Thirty-four ($n = 34$) ARGs (with >98 % nucleotide identity) were predicted in the study strain, belonging to 14 antimicrobial classes and seven different resistance mechanisms. Among these ARGs, *msrA*, *blaZ*, *blaTEM-1*, *baeR*, *oqxA*, *oqxB*, *mgrA*, *PC1_blaZ*, *OXA-437*, *blaR1*, and *sul1* were highly abundant in G1M1F genome (based on percent identity and the number of hits). In addition, the ARGs analysis revealed that 26.47 % ARGs were

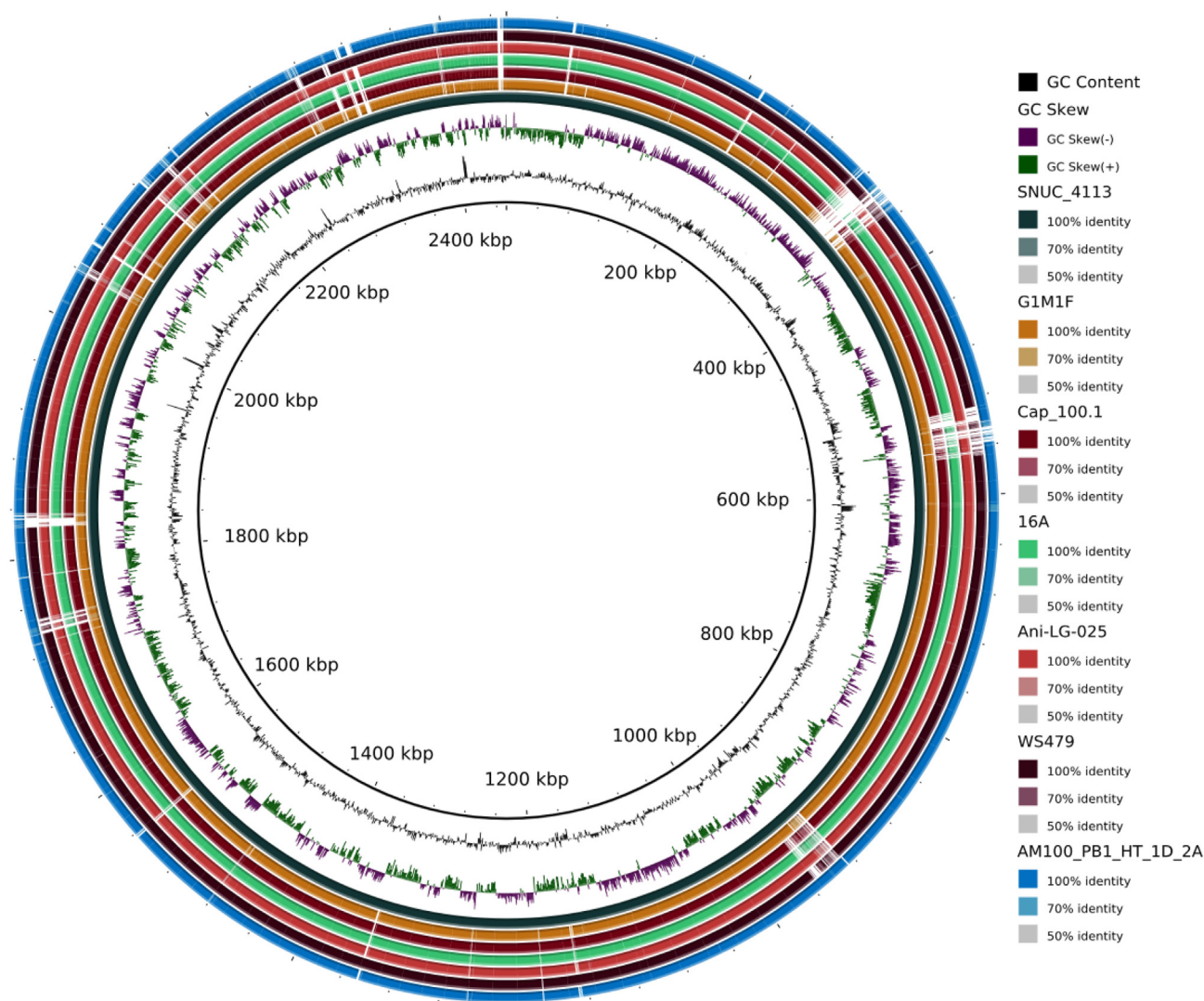


Fig. 3. Genomic characterization of *S. warneri* G1M1F. Circular representation of the *S. warneri* complete genomes. Circles (from inside to outside) 1 and 2 (GC content; black line and GC skew; purple and green lines), circle 3 (reference *S. warneri* strain SNUC_4113; dark cyan circle); circle 4 (mapped *S. warneri* G1M1F genome; carrot orange circle); circle 5 (mapped *S. warneri* Cap_100.1 genome; dark red circle); circle 6 (mapped *S. warneri* 16A genome; lime green circle); circle 7 (mapped *S. warneri* Anti-LG-025 genome; red circle), circle 8 (mapped *S. warneri* WS479 genome; dark pink circle), and circle 9 (mapped *S. warneri* AM100_PBI_HT_1D_2A genome; strong blue circle). Mapping of the genomes was done using BLASTn with an e-value cut-off 1×10^{-5} using BRIG 0.95.

found to be commonly detected by all of the four databases used (Fig. S3a). These shared ARGs included *msrA*, *mgrA*, *blaZ*, *OXA-437*, *PC1_blaZ*, *blaI*, *blaR1*, *APH(3')*, and *norC* (Fig. S3b). In this genome, the highest number of ARGs (43.75 %) were associated with multi-drug resistance (MDR; >3 antimicrobials), followed by beta-lactams (22.25 %), fluoroquinolones (13.5 %), aminoglycosides (9.5 %), macrolides (5.25 %) and sulphonamides (3.75 %) resistance. By comparing the mechanism of antimicrobial resistance of the detected ARGs, we found that more than 64 % genes were encoding for antibiotic efflux, followed by genes related to antibiotic target alteration (14.5 %), antibiotic inactivation (8.5 %), reduced permeability of antibiotics (7.4 %) and antibiotic target protection (5.6 %).

To better elucidate the virulence potentials, we identified 30 VFGs in the draft genome of G1M1F. By comparing the mechanisms of virulence, 30 % VFGs were associated with adhesion and biofilm production (e.g., *sasG*, *sasH*, *sasA*, *sasC*, *sasD*, *sasF*, *sasI*, *sasK*, and *sarA*), followed by fibronectin binding (16.67 %; *FnBPA*, *FnBPB*, *Hyp1*,

Hyp2, and *VWbp*), cellular adherence (10.0 %; *sdrC*, *sdrD*, and *sdrE*), extracellular adherence (6.67 %, *eap* and *eap_tr*), clumping factors (6.67 %; *clfA* and *clfB*) (Fig. S4a). Overall metabolic functional profile of the G1M1F genome is shown in Fig. S4b. The functional annotations showed that the draft genome contained 278 metabolic features in SEED subsystems with 34 % coverage. These subsystems were mostly represented by genes encoding for metabolisms of amino acid and derivatives (18.71 %), carbohydrates (17.60 %), protein (12.36 %), and cofactors, vitamins, prosthetic groups and pigments (7.47 %). In addition, 50 (4.29 %) genes were annotated to be responsible for virulence, disease, and defense, 36 (3.01 %) for membrane transport, 29 (2.49 %) for stress response, and 26 (2.23 %) for regulation and cell signaling (Fig. S4b). In addition, secondary metabolite-biosynthetic gene clusters (BGCs) were predicted in the draft genome (Fig. S5). The G1M1F genome possessed BGCs encoded for the biosynthesis of NI-siderophore (staphyloferrin A), terpene, opine-like-metallophore (staphylopin), and cyclic-lactone-autoinducer (Fig. S5).

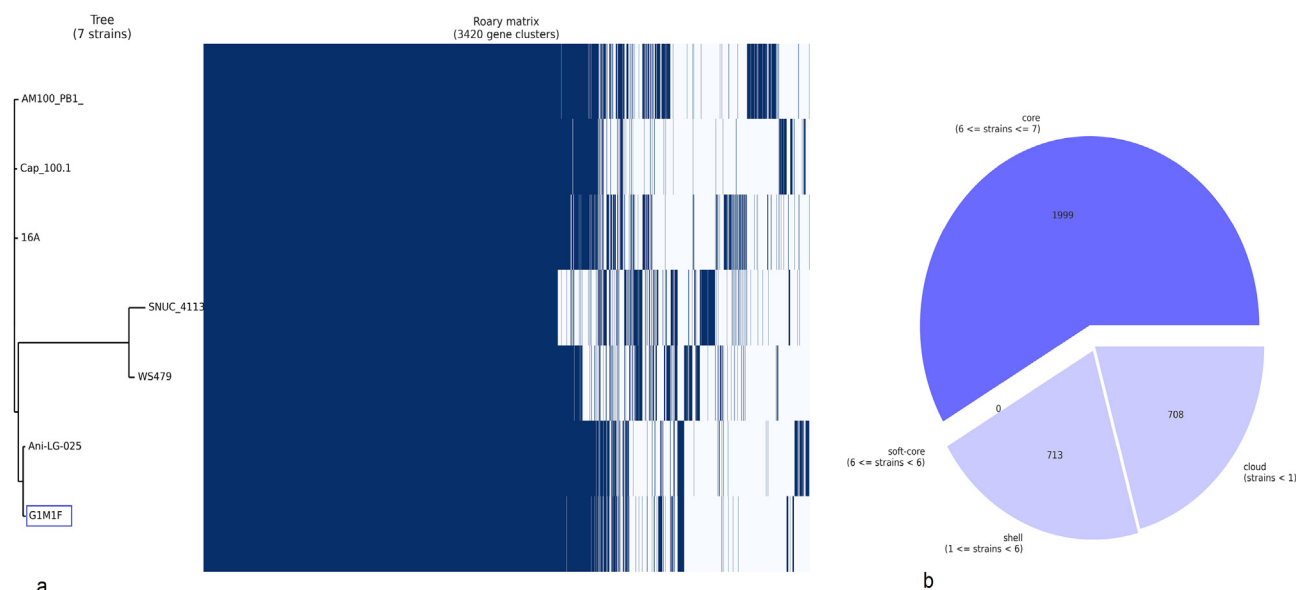


Fig. 4. Pangenome analysis. (a) Pangenome based (gene presence and absence) gene clustering matrix of G1M1F genome (enclosed in a blue box) and closely related *S. warneri* genomes. (b) Breakdown of genes in G1M1F genome. We generated the figures (a–b) based on the data obtained from Roary pangenome analysis using the roary_plots.py script.

3.4. *S. warneri* G1M1F leads to significant inflammatory lesions in the mammary gland and colon tissues of the inoculated mice

By comparing the post-infection inflammatory responses between challenged and control mice groups, we found that the G1M1F strain induced mastitis in all of the 14 mice (100.0 %) in Group-I and Group-II (Fig. 5a–e) while no clinical signs were found in Group-III mice (Fig. 5k and l). Severe gross pathological changes in mammary glands (including swollen, red and inflamed mammary glands) of mice were observed after ten days of challenge (Fig. 5c–e). In addition to mammary gland pathogenesis, inflammatory changes in the foot pads were also observed in some of the G1M1F challenged mice (Fig. 5c and d). Gross clinical signs and syndrome of mastitis pathogenesis in the challenged mice were further confirmed by histopathological examination of the mammary gland and colon tissue. Histopathological analysis showed that mice infected with G1M1F produced severe mastitis lesions characterised as scale 3 lesions, showing a profuse inflammatory infiltrate with severe tissue damage, including necrosis in the mammary gland (Fig. 5f and g) and colon (Fig. 5h and i). Mastitis associated histopathological changes in the mammary tissues included broken lobules of the mammary gland, damaged acini, destroyed or degenerated epithelial cells, with inflammatory cells, including macrophages, neutrophils, and blood cells detected in the mammary lobule, supporting connective tissue and lining of the epithelium (Fig. 5f and g). Furthermore, inflammatory changes in the colon tissues of the challenged mice included severe infiltration of inflammatory cells (e.g., macrophages, neutrophils, and blood cells) into mucosa and submucosa, degenerated mucosal structure (epithelial necrosis, extension of the subepithelial space, and structural damage of villi) (Fig. 5h and i). On the contrary, mice receiving a placebo (i.e., distilled water) showed none of such pathological changes either in the mammary gland or colon tissue, and revealed normal epithelial texture in the mammary gland (Fig. 5m) and well-arranged villi structure in the colon mucosa (Fig. 5n).

4. Discussion

Mastitis stands as one of the most significant infectious diseases in lactating mammals, with staphylococci ranking among the most common causative pathogens [1,2]. Nowadays, NAS are increasingly recognized as etiological agents associated with mastitis in dairy animals worldwide [4,9,13]. Moreover, NAS may have an impact on the colonization of *S. aureus*, a major pathogen in mammalian mastitis [13]. However, until now, no studies have explored NAS-associated mastitis in mice to better elucidate the pathophysiology of bovine mastitis. By employing both phenotypic and genomic approaches, we identified four NAS species – *S. warneri*, *S. chromogens*, *S. epidermidis*, and *S. simulans* in the feces of experimentally induced mastitis mice and shed light on their antimicrobial resistance patterns. This report for the first time uncovered unique genetic features, encompassing evolutionary phylogeny, ARGs, VFGs, and genomic metabolic potential of a NAS isolate, *S. warneri* G1M1F from the mastitis induced mice. NAS are among the most frequently isolated bacteria from dairy animals with clinical and subclinical mastitis [4,9,13]. In the current study, NAS accounted for 54.35 % of murine mastitis, of which *S. warneri* was identified as the most prevalent NAS species with a prevalence of 40.0 %. The prevalence and distribution of NAS species in mammalian mastitis are affected by several environmental and host factors, pathogens load, and antimicrobial use [4,9,13]. Similar findings on the growing importance of NAS as mastitis pathogens have been reported in bovine mastitis from different countries [4,9,13]. However, in Bangladesh, the prevalence of NAS, especially *S. warneri*, causing both bovine and murine mastitis, has not been reported so far. Therefore, the findings of this study shed light on the genomic potentials of NAS strain *S. warneri* G1M1F, which is known as a predominant pathogen in mammalian mastitis [4,9,13]. In a further experimental mouse mastitis model challenge test, the *S. warneri* G1M1F was found as a potential pathogenic NAS strain that produced marked mastitic syndromes, including both gross lesions in the mammary gland surface (e.g., swollen, red and

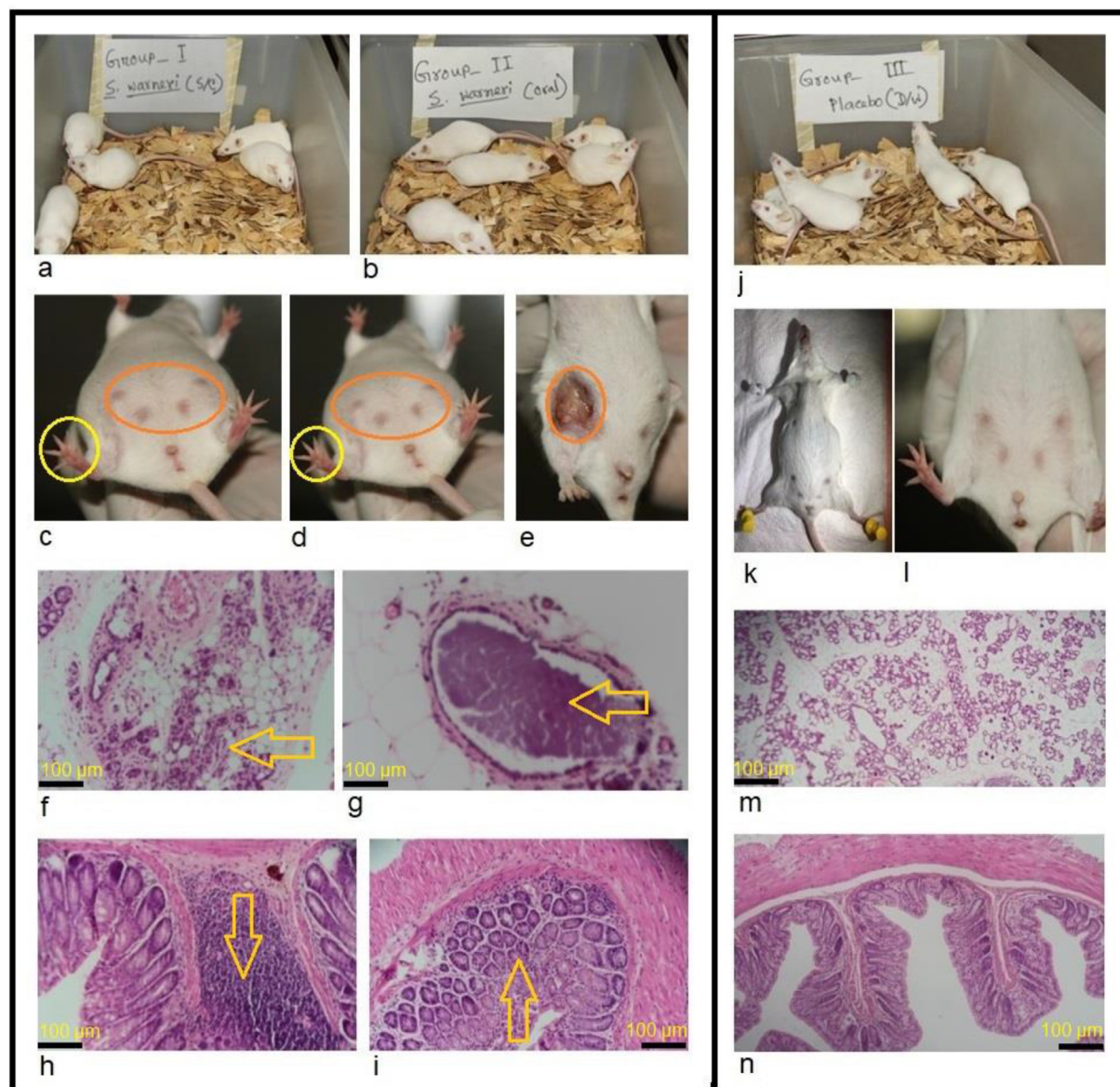


Fig. 5. Pathophysiological changes in mammary tissue and gut (colon) of mice after experimental challenge with *S. warneri* G1M1F. The mice were challenged at Day 18 of their gestation. Left panel: (a – b) Challenged mice. (a) Group - I: challenge with *S. warneri* G1M1F through subcutaneous (s/c) injection in the inguinal region; and (b) Group - II: challenge with *S. warneri* G1M1F orally. (c – d) Pathological changes in mammary glands (swollen, red, and inflamed mammary glands) of mice ten days after challenge. Inflammatory changes in the mammary glands of the mice are highlighted in orange circles. Besides mammary glands, inflammatory changes in the foot pads were also observed in some of the challenged mice (yellow circle). (f – g) Representative photomicrographs of mammary epithelial tissue after haematoxylin and eosin (HE) staining ($\times 100$ magnification). The lesions in the mammary epithelial cells are characterized by a central area of necrosis, broken lobules, damaged acini, and destroyed epithelial cells, with large numbers of inflammatory cells, predominantly neutrophils (dark bluish) and macrophages (red) (yellow arrows), in the mammary lobule, supporting connective tissue and lining of the epithelium. (h – i) Representative photomicrographs of the colon (crypts, lamina propria, muscularis mucosa, and submucosa) tissue after HE staining ($\times 100$ magnification). Histopathological changes include moderate to severe inflammatory cell infiltration into mucosa and submucosa, a disorder in mucosal structure: epithelial necrosis, extension of the subepithelial space, and structural damage of villi (yellow arrows). Scale bars: 100 μm . Right panel: (j) Control mice. Group - III: Mice received only a placebo (distilled water) orally. (k – l) Healthy mice: no visible inflammatory changes in the mammary glands. (m – n) Representative photomicrographs of (m) mammary epithelial tissue and (n) colon of healthy mice after HE staining where no inflammatory changes were observed.

inflamed mammary glands) and histopathological lesions in the mammary gland and colon tissue. The histopathological data revealed that the lobules and acini of the mammary gland were broken and damaged, many epithelial cells were destroyed and/or necrosed, and inflammatory cells were recruited to the mammary gland and colon tissue after challenge with G1M1F. These results are in accordance with the findings of our previous study [18] and several earlier reports [38]. The *in-vitro* antimicrobial assays showed that more than 68 % NAS isolates (including G1M1F) were

MDR, showing >3 antibiotics resistance properties which belongs to different classes. In this study, NAS isolates have been shown to be highly resistant to oxacillin, sulphonamides, ampicillin, nalidixic acid and gentamicin. These findings of high MDR patterns in murine mastitis associated *S. warneri* strain are in line with many of previous studies on dairy animal mastitis [13,17,18,40]. A series of previous studies reported that antimicrobial resistance against mastitis causing bacteria could vary according to the type and origin of bacteria and hosts [4,9,13]. Recently, AMR has become a

serious threat to the dairy sector of Bangladesh compromising sustainable dairy development because of irrational overuse and widespread misuse of antibiotics.

The advent of WGS for bacterial pathogens has provided a new platform to study their molecular epidemiology and potential for virulence [14,40,41]. The draft genome analysed in this study exhibited high quality genome features for analysis, with 32 contigs, 2426 protein-coding genes, and 38 RNA genes. The genome contained two prophage regions, two CRISPR arrays, and two plasmid replicons (e.g., rep5b and repUS35). Recently, these two plasmids have been detected in a methicillin resistant *S. aureus* (MRSA) strain isolated from dogs [42]. CRISPR arrays, which have been identified in many bacterial pathogens causing mastitis, play a significant role in host adaptive immune response and virulence [40,43]. One of the important findings of this study is the genomic similarities observed between the G1M1F strain and nine other NAS isolates. Pairwise comparison across these closely related NAS genomes showed 100.0 % ANI value for G1M1F. The phylogenetic placement of the G1M1F isolate aligns with several NAS associated *S. warneri* strains causing mastitis in cows, goats and buffalo cows [4,9,13]. The draft genome of G1M1F also showed a close evolutionary relationship with *S. warneri* strains isolated from human feces and blood and mouse feces. These findings corroborate with many of the previous studies which reported that NAS species are emerging pathogens of bovine and bubaline mastitis worldwide [4,9,13].

A notable discovery in this study is our capacity to accurately predict both the repertoire of ARGs and VFGs in the G1M1F genome. The *in vitro* resistance pattern of the isolate being studied aligns with the ARGs profile obtained from WGS data. We found that the majority of the ARGs were associated with MDR, indicating resistance to three or more antimicrobials. These ARGs primarily confer resistance through antibiotic efflux pumps, followed by antibiotic target alteration, antibiotic inactivation, reduced permeability of antibiotics, and antibiotic target protection mechanisms. For example, genes conferring resistance to macrolides (*msrA*), multiple drugs (*mgrA*, OXA-437), aminoglycosides (*APH(3')*), beta-lactams (*blaZ*, PC1_*blaZ*, *blaI*, *blaR1*), and fluoroquinolone (*norC*) were the predominant ARGs detected through all of the ARG databases used. Therefore, multi-drug resistance mediated by efflux pumps appears to be a widespread resistance mechanism in *S. warneri*, likely due to the unethical overuse of antibiotics in dairy animals and the extensive application of toxic chemicals and metals in agricultural settings. These ARGs can easily spread through their host bacteria to different host inhabitants of other ecosystems [44,45]. The ARGs identified in this study are concerning because the use of these antibiotics in veterinary medicine, particularly for food animals, can contribute to antibiotic resistance in humans.

Another important finding of the current study is that the draft genome of *S. warneri* G1M1F contained several VFGs associated with adhesin and biofilm production (e.g., *sasA/C/D/G/F/I/K*, and *sarA*), fibronectin binding (*FnBPA/B*, *Hyp1/2*, and *VWbp*), cellular and extracellular adherence (*sdrC/D/E*, *eap*, and *eap_tr*), and clumping factors (*clfA/B*). Of the identified VFGs, *sasH* was significantly associated with invasive staphylococcal disease in humans [46] whereas surface protein G (*sasG*) was reported to promote biofilm production [47]. Fibronectin binding proteins A and B (*FnBPA* and *FnBPB*) mediate the adhesion of *S. aureus* to fibrinogen, elastin and fibronectin. These genes have been detected in human and bovine originated *S. aureus* strains that are associated with immune evasion mechanisms in different diseases [48]. Another important VFG, Von Willebrand factor-binding protein (*VWbp*), is a leading adhesion protein in staphylococcal pathogenesis [49]. Adhesion is a crucial step in staphylococcal colonization of the host and occurs before cell penetration, internalization, and chronic infection [50].

The development of biofilms aids in the evasion of host antimicrobial components and facilitates the exchange of genetic material [17,51]. Furthermore, recently both Szczuka et al. [52] and Louail et al. [53] have reported evidence that *S. warneri* exhibited the ability to invade epithelial cells. Therefore, detected VFGs may play a crucial role in mastitis-causing *S. warneri* by helping them evade host defenses and biofilm production, and successfully colonize the udder and/or mammary gland. However, the specific mechanisms through which these VFGs contribute to microbial colonization in the udder and/or mammary gland are not well understood. Remarkably, our data strongly suggest that adhesion and binding abilities, enterobactin production, and multiple antibiotic resistance through efflux pumps are essential genomic attributes enabling the *S. warneri* strain to induce mammary gland inflammation.

We also sought to predict metabolic functional genes and pathways within the genome of *S. warneri* G1M1F. Our analysis identified several important genes and proteins in the G1M1F strain that are associated with different subsystem categories and metabolic functions, supporting their implications with mastitis pathogenesis [40,41,54]. The *S. warneri* G1M1F strain possessed a higher number of genes involved in the metabolism of amino acids and derivatives, carbohydrates, proteins, cofactors and vitamins, stress response, membrane transport, regulation and cell signaling, as well as virulence, disease, and defense. Previous research has indicated that bacterial metabolites play a role in modulating host immune functions and disease pathophysiology [12,13]. Additionally, we identified several gene clusters involved in secondary metabolite biosynthesis in the G1M1F genome. These secondary metabolites could potentially serve as a valuable source of novel bioactive compounds. Previous studies have suggested that bacterial secondary metabolites could play a role in their pathogenicity [55], but the specific functions may differ depending on the type and quantity of these bioactive compounds. Hence, further research is warranted to investigate the secondary metabolites produced by *S. warneri* strains and their specific pathogenic properties in mastitis.

One notable limitation of our study lies in the relatively small sample size used for WGS analysis. Additionally, the antimicrobial susceptibility and MIC tests focused solely on antibiotics, neglecting potential alternatives such as staphylococcal bacteriolysin (e.g., lysostaphin) and nanoparticles/metals, which could prove effective against *S. warneri* G1M1F. To enhance the robustness of our findings, it is imperative to broaden the scope of research by augmenting the sample size and diversifying the animal population, considering factors such as breed, parity, body condition, and lactation stages. Expanding the sample size not only addresses the limitations mentioned but also facilitates a more comprehensive understanding of the prevalence of NAS species. It enables a thorough exploration of their ARG repertoire, virulome, and potential genomic mechanisms implicated in mastitis pathogenesis. By overcoming these limitations, we can significantly contribute to the development of targeted and effective strategies for managing mastitis in dairy animals. This, in turn, holds the promise of improving overall animal health and mitigating the risk of antimicrobial resistance.

5. Conclusions

This study has unveiled significant genomic characteristics in a multidrug-resistant non-aureus staphylococci (NAS) strain, exemplified by *S. warneri* G1M1F, isolated from the gut of a mouse with experimentally induced mastitis. *S. warneri* G1M1F demonstrated a diverse antimicrobial resistance pattern, featuring an array of antimicrobial resistance genes (ARGs) that conveyed resistance

through various mechanisms, culminating in a multidrug-resistant (MDR) genotype. Phylogenetic and comparative genome analyses revealed genetic similarities between the G1M1F strain and other *S. warneri* strains associated with mastitis in dairy animals, originating from human feces and blood across diverse geographical regions. Notably, the G1M1F strain harboured several crucial virulence factor genes (VFGs) implicated in adhesion and biofilm production. The experimental induction of mastitis in mice challenged with this NAS strain, particularly *S. warneri* G1M1F, resulted in significant inflammatory changes in both mammary gland and colon tissues. These findings underscore the emergence of a novel NAS strain with genomic traits linked to mammalian mastitis. The findings of our study suggest the ongoing evolution and adaptation of bacterial pathogens in mammalian mastitis, emphasizing the need for vigilance and continued research into emerging strains that could impact animal health. Further investigations are warranted to elucidate the varying virulence strategies employed by gut-associated *S. warneri* strains in mastitis pathogenesis across different hosts and demographics. Additionally, our findings bear implications for the development of new strategies for mastitis prevention, treatment, and control.

Data availability

The assembled whole genome sequence of the *S. warneri* G1M1F has been deposited in GenBank under accession no. JAQPC000000000.1, and the assembly reports of the genome are also available under GenBank assembly accession no. GCA_028464135.1. The Illumina whole genome sequence data are available in the National Center for Biotechnology Information (NCBI) under BioProject accession number PRJNA929689 and BioSample accession no. SAMN32968848. The version described in this paper is the first version.

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Authors contribution

M.N.H. conceptualized and designed the study, provided resources and acquired data. M.N.H., G.M. F. and Z.C.D. conducted experiment, analysed data, interpreted results, and drafted the original manuscript. M.N.H., T.I.S., M.A.M., M.A.H. and T.I. critically reviewed and edited the manuscript. All authors approved the final version of the article.

Declaration of competing interest

The authors declare no conflicts of interest with the contents of this article.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.micinf.2023.105285>.

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